

of LLMs: *how well they adapt and remain accurate as new data emerges and contexts evolve?* As an example of using LLMs to predict political outcomes, we gather new factual data about future events, such as "What will President Biden's approval rating be as of 7 June 2024, according to FiveThirtyEight?". To ensure objectivity in our predictions, answers are typically formatted as ranges within a prediction market scenario. For instance, President Biden's approval rating might be categorized into intervals such as lower than 34%, at least 34.0%, but less than 36.0%, at least 36.0%, but less than 38.0%, up to 46.0% or higher.

However, since collecting actual future data is impossible, we adopt a retrospective study approach. We use data available after the event's question was posed and before the prediction market's closure, treating this as "future" data for model evaluation. This ongoing, online evaluation objectively assesses the model's predictive capability. We define prediction accuracy, denoted by Acc , as follows:

$$Acc = \frac{N_{\text{correct}}}{N_{\text{total}}} \quad (2)$$

where N_{correct} is the number of correct predictions among N_{total} future prediction questions. See Sec. 4.1 for the collected future prediction question set over time. This metric evaluates the effectiveness of LLMs in predicting future events. This approach also allows continuous evaluation with the latest data, ensuring the assessment reflects the model's current capabilities and the evolving information landscape.

3 Temporal Generalization through the Lens of Compression Intelligence

This section discusses how various models handle data from different periods, showcasing their performance stability or volatility.

Key Findings

- In general, while the ranking of models varied slightly between datasets, the performance of individual models remained relatively stable over time within each dataset type, as can be seen in Figure 3 and 4. Here we made a discovery regarding the distinct characteristics of the Wikipedia and BBC datasets at 3.1.2.
- Another interesting observation was made in the arXiv dataset shown in Figure 5, where

a sharp decrease in BPC occurred around March 2024. This timing coincides with the peak in ChatGPT usage, as reported by Exploding Topics², which saw over 1.8 billion visits in that month. In essence, it appears that humans may be adapting their writing to more closely resemble LLM outputs, rather than LLMs significantly enhancing their ability to model human language.

3.1 Temporal Bias

3.1.1 Protocol for Quantitative Measurement

Data sources Our system aggregates data from various platforms, including Yahoo for financial news, BBC for political insights, Reddit for discussions, Wikipedia for encyclopedic updates, arXiv for academic research, GitHub for software trends, and Quora for diverse questions and answers. Utilizing a Python-based crawling framework, it's designed for adaptability and efficient data collection, with Playwright addressing navigation and specific site challenges. We also did some basic data pre-processing, due to space limitation, details in Appendix F. An overview of the data volume and the average entry length from a single crawl is provided in Appendix G. Experiment settings are listed in Appendix A.

3.1.2 Observation of Temporal Bias in Compression Intelligence

To comprehensively explore the temporal generalization capabilities of models, we plot the BPC curve over time.

In text data before release, a negative slope indicates the presence of **Neophilia Bias**. Conversely, a positive slope would suggest that the model's performance is decreasing, which shows **Nostalgia Bias**.

ArXiv BPC has a huge decline after GPT is widely used, as mentioned in Key Findings. Models show stronger **Nostalgia Bias** when evaluated with BBC text than Wiki text. From the BPC charts, it's not easy to spot differences between models. However, our calculations in Appendix I reveal variations in how well each model can adapt specific text over time.

Difference of Text Feature An insightful reviewer helped us find the reason of the different patterns of Wiki and BBC text. New Wiki texts with

²<https://explodingtopics.com/blog/chatgpt-users>

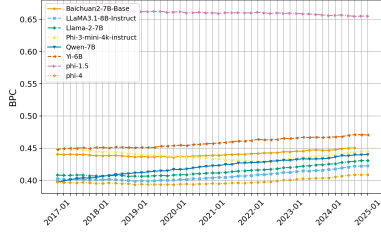


Figure 3: BPC on Wiki text

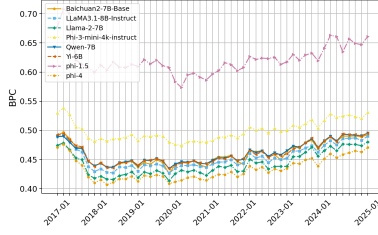


Figure 4: BPC on BBC news text

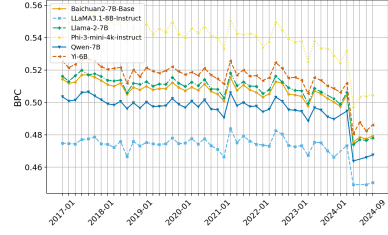


Figure 5: BPC on arXiv text

recent temporal markers do not necessarily imply that the content is new in a temporal context. For instance, in 2017-01, there was a Wiki article titled "History of India". However, the information within it was not new and could potentially have been sourced from other materials. Conversely, BBC texts are news articles and are highly time-sensitive.

To quantify the difference, we uniformly sampled approximately 500 data pieces from both the Wiki and BBC datasets (taking into account that news may also cover historical events) and asked gpt-4o to classify the content. The results are as follows:

- Wiki: Historical: 97.83%, Others³: 2.17%.
- BBC: Recent: 91.86%, Historical: 8.14%.

So, it is reasonable that Wiki text has a flat curve in BPC.

Claim the Value of Wiki Dataset Since the Wikipedia dataset is continuously updated, it is not easily subject to direct contamination. Although a substantial portion of Wiki pertains to old information, this content still represents valuable knowledge that we need. An ideal model should consistently perform well on such knowledge. (If a model shows a decline in performance on Wikipedia, it is likely that the model is overfitting.)

3.2 Temporal Degeneration

3.2.1 Protocol for Quantitative Measurement

Definition 3.1. Temporal Bias Index (TBI) For each model in a certain period, we record BPC_i at n time points, forming n data points (i, BPC_i) , where $i = 1, \dots, n$. Here the time interval between BPC_i and BPC_{i+1} is set as two month. We then fit a linear model:

$$BPC_i = a * i + b \quad (3)$$

³"Others" refers to content not associated with specific historical events, focusing instead on general, educational, and ever-relevant information.

then Temporal Bias Index (TBI) is defined as the coefficient a . TBI is crucial for understanding **the direction and rate of change in model performance over time**.

To quantify model temporal generalization, we computed the BPC for various models on the Wiki and BBC datasets and evaluated their performance using this index. Specifically, We established a base BPC using data from the six months prior to each model's release. After that, we measured the changes in BPC at 3, 6, 9 and 12 months post-release relative to the base BPC; and assessed the overall trend in model performance using TBI. The specific results are detailed in Appendix H.

3.2.2 Observation

Temporal generation pattern is mostly determined by the nature of data Models perform stably on Wikipedia but show more variation on BBC. This reveals their ability to handle well-structured and formal text like Wiki and insufficient adaptability to dynamic text like BBC text. The overall BPC for BBC texts shows an increasing trend, indicating gradual performance degeneration.

Individual models show considerable differences in performance on new data; for instance, Yi, and Baichuan2 series adapt relatively well to new texts, while Phi-1.5 and OPT series exhibit poorer generalization (see Appendix I). These findings indicate that in dynamic environments, different models have significantly varying adaptation capabilities.

Model performance positively correlates with performance decay. (see Tab. 1). There is a positive correlation between model attributes (such as model size and release date) and performance decay. A positive BPC slope indicates an increase in BPC over time, reflecting performance degeneration.

In the Wiki dataset, larger models deteriorate faster (correlation coefficient of 0.504), suggesting a trade-off between capability and flexibility.

Table 1: Correlation between Model Attributes and Their TBI on Wiki and BBC text

Correlation	Model Size	Release Date	MMLU	Humanities	Social-Sciences	STEM	MMLU Other	GSM8	Ar20c	Hella-wasg	Trust-fulQA	Wino-grande	Long-bench
Wiki	0.504	0.267	0.599	0.500	0.594	0.393	0.528	0.473	0.380	0.581	0.200	0.632	0.404
BBC	-0.044	0.313	0.384	0.317	0.311	0.186	0.153	0.188	-0.071	0.248	0.485	0.204	0.549

In more dynamic environments (such as the BBC dataset), the impact of model size on performance decay is smaller. Additionally, newer models show higher correlations, meaning that despite their advanced capabilities, they may deteriorate faster. These results emphasize the importance of designing models that can maintain robustness over time and across various data environments. As machine learning applications grow in complexity and scale, temporal generalization will become increasingly critical.

4 Temporal Generalization through the Lens of Future Event Prediction

Advantages of Future Event Prediction over Compression Intelligence Compression Intelligence is limited by its inapplicability to closed-source models, where key indicators like logits or losses are inaccessible. This exclusion narrows comparative analysis and hinders a comprehensive understanding of the language technology landscape. Additionally, language likelihood can be biased by models finely tuned to specific formats or domains, skewing overall evaluation and presenting a distorted view of a model’s general performance.

This motivated us to research temporal generalization and bias from the perspective of future prediction.

In this section, we investigate temporal generalization (in Sec. 4.1) and biases (Sec. 4.2) through the Lens of Future Event Prediction.

Key Findings

- Models prefer older knowledge, with non-uniform knowledge distribution skewed towards the past. This poses a risk as users expect up-to-date knowledge close to the model’s release date.
- Within model series such as Claude, Gemini, and GPT, newer versions are becoming more powerful but also experience more rapid performance decline. Miniature models signif-

icantly underperform compared to their full-sized counterparts in temporal tasks.

- Newer models generally exhibit stronger performance. Closed-source models tend to perform slightly better at the beginning, yet open-source models demonstrate superior generalization capabilities.

4.1 Temporal Biases

4.1.1 Protocol for Quantitative Measurement: a Hypothesis Testing Method

We discussed generally what is past, present, and future for a model in Sec 2.1.1. Now, let’s clarify the context of the prediction problem. For each prediction task, there is an initiation time (when the question is posed) and a close time (when the answer is revealed). If a question’s closure time precedes the model’s release date (for instance, by more than three months), then that question is considered part of the *past* for that model. The *present* refers to questions whose close time is near the release time of the model, capturing the period around when the model was launched.

To investigate the two mentioned temporal biases in Sec. 2.1.2, we propose two hypothesis testings as follows. These tests help practitioners make informed decisions despite the randomness inherent in the question datasets.

Hypothesis Testing 1. Neophilia Bias:

$$\begin{aligned} T_1 : \quad & H_0 : \mathbf{E}[acc_{past}] = \mathbf{E}[acc_{present}], \\ & H_1 : \mathbf{E}[acc_{past}] < \mathbf{E}[acc_{present}] \end{aligned} \quad (4)$$

This hypothesis tests if there is a significant increase in model accuracy for current questions compared to past questions, suggesting a bias towards newer content. The p-value $\hat{p}$ quantifies the strength of evidence against the null hypothesis.

Hypothesis Testing 2. Nostalgia Bias:

$$\begin{aligned} T_2 : \quad & H'_0 : \mathbf{E}[acc_{past}] = \mathbf{E}[acc_{present}], \\ & H'_1 : \mathbf{E}[acc_{past}] > \mathbf{E}[acc_{present}] \end{aligned} \quad (5)$$

Conversely, this hypothesis tests if there is a significant increase in model accuracy for past questions, indicating a preference for older or historical